

Online Voting System - Final Project Report

Database Systems

Course: CL2005 – Database Systems Lab

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University: FAST-NUCES CFD Campus

Group Members:

- 24F-0773
- 24F-0671

Domain Description

The Online Voting System is a database-driven application designed to make elections digital, secure, and efficient. Using this system, voters can cast their votes online, candidates can be managed easily, and election results are generated automatically.

Traditional voting systems involve manual processes which are:

- Slow
- Error-prone
- Less transparent

Therefore, we developed a centralized database system to improve efficiency and reliability.

Manual voting systems face several problems:

- Risk of fake or duplicate voting
- Slow result calculation
- Difficult data management
- No real-time statistics

The goal of this project is to solve these issues using a database system.

Objectives

Main objectives of the project are:

- Build a secure voter registration system
- Manage elections, candidates, and parties
- Prevent duplicate voting
- Automatically count votes and generate results
- Provide an admin dashboard with statistics

Requirements

Functional Requirements

The system must provide the following features:

User / Voter

- Can register and login
- Can view active elections
- Can cast a vote (only one vote per election)

Admin

- Can add, edit, and delete parties
 - Can manage candidates
 - Can create elections
 - Can view results
 - Can view dashboard statistics
-

Data Dictionary

Table	Description
USERS	Stores all users (admins and voters)
ADMIN	Stores admin specific information
VOTER	Stores voter specific information
PARTY	Stores political parties
CANDIDATE	Stores election candidates
ELECTION	Stores election details

Table	Description
VOTE	Stores votes cast by voters
RESULT	Stores election results

Database Design

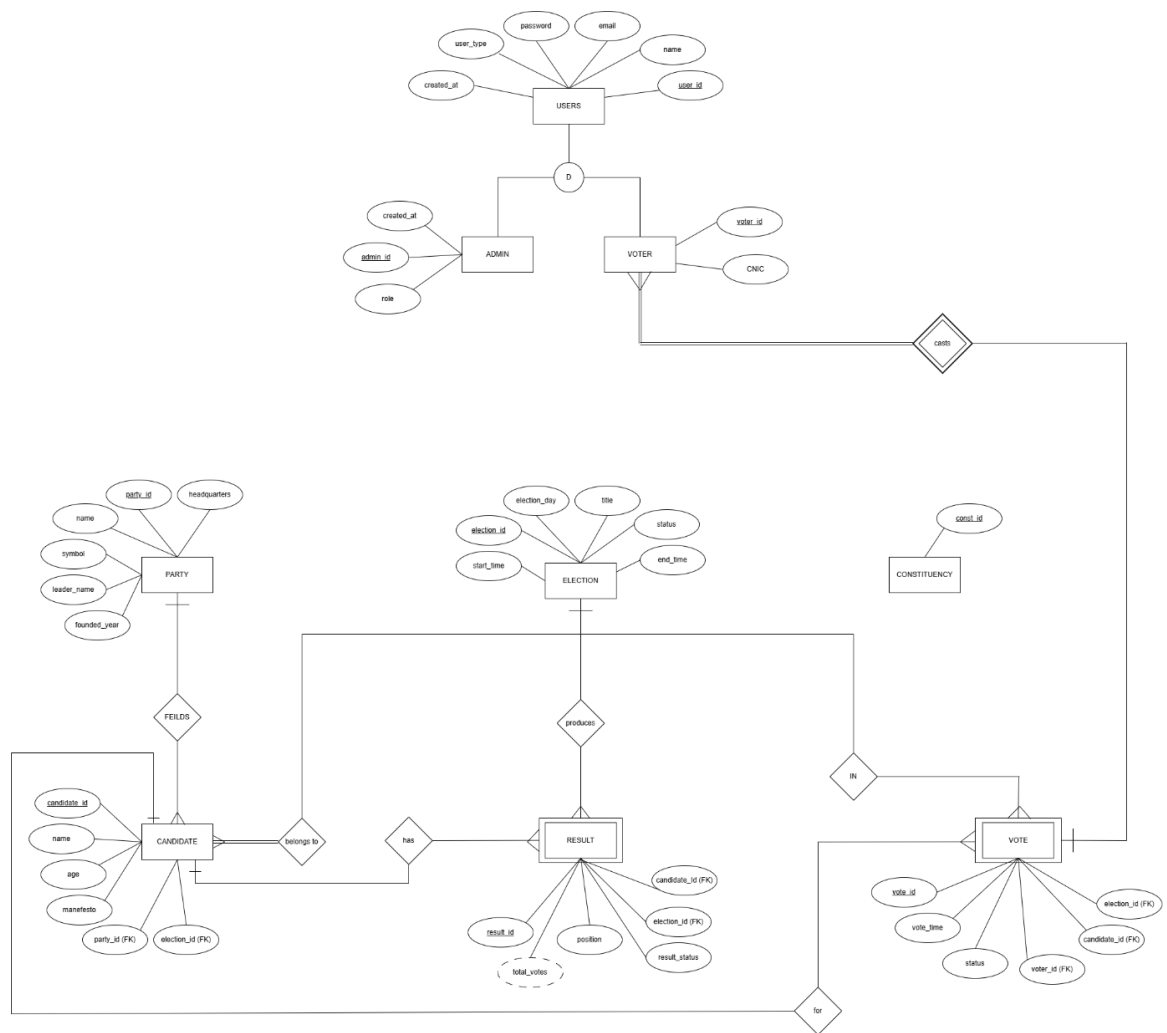
EERD Description

The system includes the following relationships:

- One **Party** can have many **Candidates** (One-to-Many)
- One **Election** can have many **Candidates**
- One **Voter** can cast only **one vote per election**
- **Results** are generated from the votes

Key relationships:

- USERS → ADMIN / VOTER (specialization)
- PARTY → CANDIDATE
- ELECTION → CANDIDATE
- VOTER → VOTE → CANDIDATE



Normalization

The database is normalized up to **Third Normal Form (3NF)**.

1NF

- No repeating groups
- All attributes are atomic

2NF

- Removed partial dependencies

- Broke composite key dependencies

3NF

- Removed transitive dependencies
Example: Party information is stored in a separate PARTY table instead of the CANDIDATE table.

Schema Description

USERS

Stores all system users.

Attributes:

- user_id (Primary Key)
- name
- email
- password
- user_type (ADMIN/VOTER)
- created_at

PARTY

Stores political parties.

Attributes:

- party_id (Primary Key)
- party_name
- symbol

CANDIDATE

Stores election candidates.

Attributes:

- candidate_id (Primary Key)
 - name
 - party_id (Foreign Key)
 - election_id (Foreign Key)
-

ELECTION

Stores election details.

Attributes:

- election_id (Primary Key)
 - election_name
 - start_date
 - end_date
-

VOTE

Stores votes cast by voters.

Attributes:

- vote_id (Primary Key)
- voter_id (Foreign Key)
- candidate_id (Foreign Key)
- election_id (Foreign Key)

Constraint:

One voter can vote only once per election.

RESULT

Stores election results.

Attributes:

- result_id (Primary Key)
 - candidate_id (Foreign Key)
 - total_votes
-

Query

Query 1 – Register User

Purpose: Insert a new voter.

```
INSERT INTO USERS(name,email,password,user_type)
VALUES('Ali','ali@gmail.com','123','VOTER');
```

Query 2 – Add Candidate

```
INSERT INTO CANDIDATE(name,party_id,election_id)
VALUES('Ahmed',1,1);
```

Query 3 – Cast Vote

```
INSERT INTO VOTE(voter_id,candidate_id,election_id)
VALUES(2,3,1);
```

Query 4 – Calculate Results

```
SELECT candidate_id, COUNT(*) AS total_votes
FROM VOTE
GROUP BY candidate_id;
```

PL/SQL

```
CREATE TRIGGER trg_one_vote
ON VOTE
FOR INSERT
AS
IF EXISTS(
  SELECT * FROM VOTE v
  JOIN inserted i
```



```
ON v.voter_id=i.voter_id AND v.election_id=i.election_id
)
BEGIN
RAISERROR('Voter already voted!',16,1);
ROLLBACK;
END;
```

GUI

ONLINE VOTING SYSTEM

☐ Voter

☒ Admin

Enter Email

Enter Password

Login

INSTRUCTIONS

- > Log in using your registered CNIC/Email and Password.
- > Each voter can cast only one vote.
- > Verify your details before voting.
- > Select your candidate carefully.
- > Click "Submit Vote" to confirm (cannot be changed later).
- > Do not share your login credentials.
- > Logout after completing your vote.

*Note: Your vote is secure and confidential.

Form2

Admin Dashboard

Add Election

Add Party

Add Candidates

View Results

Logout

Stats

Total votes
10

Total voters
18

Total Elections
10

Winner
Ali Raza

PartyForm

Add Party

Enter party

Enter leader Name

Enter Headquarter

Add

Symbol

Enter year

Load

	party_id	name	symbol	leader_name	founded_year
▶	1	PTI	Bat	Imran Khan	1996
	2	PML-N	Lion	Nawaz Sharif	1993
	3	PPP	Arrow	Bilawal Bhutto	1967
	4	MQM	Kite	Khalid Maqbool	1984
	5	ANP	Lantern	Asfandiyar Wali	1986

ADD CANDIDATE

Enter name

Age

Manifesto

Party

Election

ADD

TERMS & INSTRUCTIONS

> Enter accurate candidate details (Name, Party, Constituency).

> Ensure the candidate belongs to the selected constituency.

> Avoid duplicate entries for the same candidate.

> All required fields must be filled correctly.

> Only authorized admins can add candidates.

> Verify information before saving.

*Note: Incorrect or duplicate entries may be removed by the system.

ElectionForm

ADD ELECTIONS

Election Name

Election Date

Wednesday, May 6, ▾

Start Date

Wednesday, May 6, ▾

End Date

Wednesday, May 6, ▾

Status

▾

Add

TERMS & CONDITIONS

> Enter accurate election details (Name, Date, Constituency).

> Election name must be "unique".

> Dates must be valid and not in the past.

> Ensure constituency is correctly assigned.

> Once created, critical details should not be changed.

> Only authorized admins can add elections.

> Any false or duplicate entry may lead to removal.

*Note: By adding an election, you agree to follow system rules and ensure data integrity.

ResultForm

Result

	candidate_id	CandidateName	party_id	election_id	TotalVotes
▶	1	Ali Raza	1	1	2
	2	Ahmed Gillani	2	1	2
	3	Zain Asif	3	1	2
	4	Usman gulzar	4	1	2
	5	Bilal Shafi	5	1	2
	6	Kashif Nazir	1	2	0
	7	Adnan Khan	2	2	0
	8	Salman Raof	3	2	0
	9	Rizwan Randhawa	4	2	0
	10	Farhan Khan	5	2	0
	11	Imran Tariq	6	3	0

VotingFrom

CAST YOUR VOTE

Election

Candidate

☐ I've read all the instructions

☐ I agree to the given terms and conditions.

Vote

TERMS & INSTRUCTIONS

> Select your constituency and view available candidates.

> Choose "one candidate only".

> Review your selection carefully before submitting.

> Click "Cast Vote" to confirm your choice.

> Vote cannot be changed once submitted.

> Each voter is allowed "one vote only".

> Do not refresh or close the form during submission.

*Note: Your vote is secure, confidential, and final.

VoterFrom

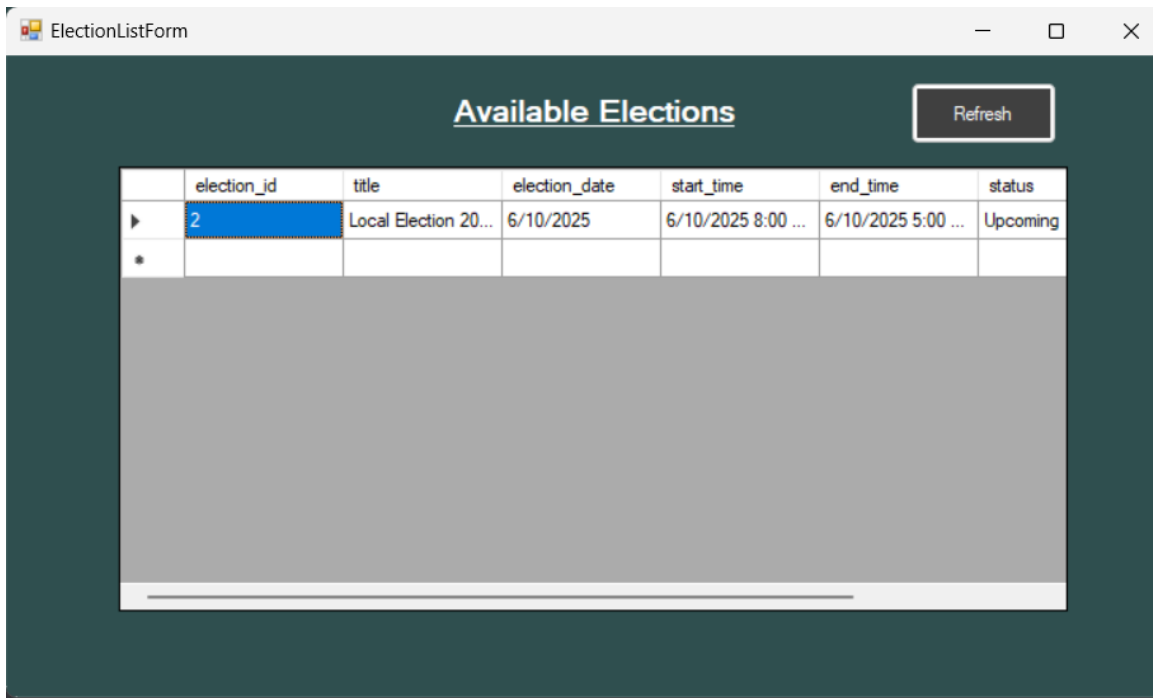
Voter Dashboard

View Elections

Cast Vote

View Results

Logout



Dashboard

Dashboard shows the following metrics:

- Total Users
- Total Voters
- Total Candidates
- Total Elections
- Total Votes Cast

This helps the admin monitor the system in real-time.

Challenges

- Implementing SQL constraints

- Preventing duplicate voting
- Connecting forms with database
- Fixing DataGridView data loading issues

Lessons

- Real-world database design experience
- SQL Server with C# integration
- Use of triggers and constraints
- Teamwork and debugging skills